NASA Space Apps Challenge 2026 — Official Technical Proposal

FLARE-CHRONOS

Fire Radiative Power Longitudinal Alignment, Radiometric Equalization & Cross-Sensor Harmonization for Multi-Decadal Observations

Physics-Guided Cross-Sensor Harmonization & 26-Year Unified Burning Activity Calendar (`2000–2026`)

Challenge: *Harmonization of MODIS and VIIRS Hot Spots (NASA Earth Science Division / LANCE FIRMS)*
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1. Executive Summary

For more than a quarter-century (`2000–2026`), NASA's **MODIS** (*Moderate Resolution Imaging Spectroradiometer* aboard **Terra** and **Aqua**) and **VIIRS** (*Visible Infrared Imaging Radiometer Suite* aboard **Suomi-NPP**, **NOAA-20**, and **NOAA-21**) have served as humanity's primary planetary fire-monitoring constellation through **NASA LANCE FIRMS** (*Fire Information for Resource Management System*).

However, as **Terra**, **Aqua**, and **Suomi-NPP** approach end-of-service in `2026`, climate scientists, forestry ministries, and wildfire incident commanders face a critical scientific barrier: **naive aggregation of MODIS (`1 km`) and VIIRS (`375 m`) hotspot records creates severe artificial regime shifts (`+380%` to `+1,311.2%`)** whenever a new satellite enters the constellation (`2012` Suomi-NPP, `2018–2020` NOAA-20, `2024` NOAA-21). Conversely, discarding MODIS before `2012` destroys the first `12` years of the satellite climate record (`2000–2011`).

FLARE-CHRONOS solves this challenge from first principles of orbital mechanics, spherical scan geometry, and infrared radiometry:

- **Physical & Statistical Cross-Sensor Harmonization (`packages/harmonization-core`)**: Instead of treating hotspots as dimensionless points, FLARE-CHRONOS reconstructs each sensor's exact elliptical ground footprint ($\Delta S(\theta) \times \Delta T(\theta)$), allocates radiative energy onto **Uber H3 Resolution 7 (`5.161 km²`) and Resolution 8 (`0.737 km²`)** equal-area hexagons via 13-point spherical quadrature, unifies astronomical Solar Zenith Angles ($SZA < 85^{\circ}$), filters South Atlantic Anomaly (SAA) cosmic-ray spikes (`4.63%` of nighttime S-NPP detections over South America), calibrates Fire Radiative Power (FRP) across $N = 20{,}786$ co-incident Aqua–SNPP cells (reducing Kolmogorov-Smirnov

distributional divergence D_{KS} from `0.1153` to `0.0353`, a `69.4%` **error reduction**), clusters multi-satellite afternoon overpasses within $\tau = 60$ **min** into independent **Fire Episodes**, and normalizes by effective observation effort $\Omega(y, \varphi)$.

- **Elimination of the `+1,311.2%` Multi-Sensor Mirage**: Validated on `1,609,182` **real NASA FIRMS detections across `134` cached CSV slices** (`451,488` detections in the canonical 5-day peak windows), FLARE-CHRONOS eliminates the artificial `+1,311.2%` **step-jump** in the Mekong Subregion & Vietnam (`2024` raw index `1,411.2` vs. `2005 = 100.0`), recovering the true physical index of `90.5` **(`-9.5%` below `2005`)** while preserving true historical **El Niño** drought peaks in `2010` (`184.9`) and `2016` (`133.0`).
- **26-Year Harmonized Burning Activity Calendar (`26 Years × 73 Pentads`)**: Produces a continuous `2001–2026` matrix (`25` **Measured Epochs** `2001–2025` + `1` **Interpolated Year** `2026` per region) of Harmonized Fire Density (**HFD**), Harmonized Radiative Flux (**HRF**), and Standardized Fire Anomaly Index (**SFAI** Z-scores computed strictly from Measured Epochs per ADR-0010).
- **Dual-System Cognitive AI Harness (`apps/ai-service`) & Interactive Workbench (`apps/workbench`)**: Pairs a LangGraph **System 1 Reflexive Layer** + **System 2 Deliberative Layer** across `15` **Use Cases (`5` System 1 + `10` System 2)** and `18` **registered deterministic** `@tool` **functions** (`19` total with the internal CSV tool), backed by a `VerifiedFactsLedger` , a `NumericGroundingVerifier` , an identity-bound JWT **Human Sign-Off Gate**, and a live **Next.js 16 + Three.js 3D Globe +** `d3-geo` **Equal-Area SVG + H3 Workbench**.

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2. Official NASA Space Apps Submission Form Packet

This section provides the exact structured fields required by the **NASA Space Apps Challenge 2026 Project Submission Form** for Global Judging.

Submission Form Field	Official Response for NASA Space Apps Judges
Project Name	**FLARE-CHRONOS** (*Fire Radiative Power Longitudinal Alignment, Radiometric Equalization & Cross-Sensor Harmonization for Multi-Decadal Observations*)
Challenge Addressed	**Harmonization of MODIS and VIIRS Hot Spots** *(NASA Earth Science Division / LANCE FIRMS)*
High-Level Project Summary — What did you develop?	We developed **FLARE-CHRONOS**, an open-source physics-guided cross-sensor harmonization engine, a **26-Year Harmonized Burning Activity Calendar (`26 Years × 73 Pentads` , `2001–2026`)**, a **Dual-System LangGraph Cognitive AI Harness (`15` Use Cases,** `18` **registered deterministic** `@tool` **functions)** with human-in-the-loop sign-off, and an interactive **Next.js 16 + Three.js 3D Globe +** `d3-geo` **Equal-Area SVG + Uber H3 Workbench**.
High-Level Project Summary — How does it address the challenge?	It resolves the five physical discontinuities between **MODIS (`1 km`)** and **VIIRS (`375 m`)**—bow-tie scan-angle footprint expansion (**74.23** × MODIS-edge to VIIRS-nadir area ratio), minimum **FRP** detection-limit divergence (**41.20 MW** vs. **3.40 MW**), multi-satellite afternoon overpass double-counting (`Aqua` + `S-NPP` + `NOAA-20` + `NOAA-21`), NRT/Standard day-night flag inconsistency (**SZA** < **85°**), and South Atlantic Anomaly (**SAA**) cosmic-ray false alarms—producing a homogeneous VIIRS-equivalent fire climate record from `2000` to `2026` .

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Submission Form Field	Official Response for NASA Space Apps Judges
High-Level Project Summary — Why is it important?	Without harmonization, adding `Suomi-NPP`, `NOAA-20`, and `NOAA-21` inflates raw hotspot counts by `+1,311.2%` in the Mekong subregion and `+714.3%` in the Amazon, misleading policymakers into confusing constellation expansion with wildfire growth. As Terra, Aqua, and Suomi-NPP retire in `2026`, FLARE-CHRONOS preserves the continuity of humanity's 26-year planetary fire record for climate science and early-warning operations.
Space Agency Data & Resources Used	**1. NASA LANCE FIRMS Active Fire Archives (`1,609,182` real detections across `134` cached CSV slices):** • `MODIS_SP` & `MODIS_NRT` (Collection 6.1 `MCD14ML` / `MCD14DL`, Terra & Aqua, `1 km`, `2000–2026`) • `VIIRS_SNPP_SP` & `VIIRS_SNPP_NRT` (Collection 2 `VNP14IMGML` / `VNP14IMGTDL_NRT`, Suomi-NPP, `375 m`, `2012–2026`) • `VIIRS_NOAA20_SP` & `VIIRS_NOAA20_NRT` (`VJ114IMGML` / `VJ114IMGTDL_NRT`, NOAA-20 / JPSS-1, `375 m`, `2018–2026`) • `VIIRS_NOAA21_SP` & `VIIRS_NOAA21_NRT` (`VJ214IMGTDL_NRT`, NOAA-21 / JPSS-2, `375 m`, `2024–2026`) **2. NASA LANCE FIRMS REST APIs:** `data_availability` & `area/csv` endpoints. **3. ERA5 Reanalysis Cloud-Cover Climatology:** cached under `data/cloud/` for observation effort $\Omega(y, \varphi)$.
Third-Party Materials & Open-Source Licenses	**Scientific Literature:** Giglio et al. (2016) *RSE*; Schroeder et al. (2014) *RSE*; Wolfe et al. (2013) *JGR*; Wooster et al. (2005) *JGR*; Freeborn et al. (2014) *GRL*; Sofiev & Kouznetsov (2026) *ACP*. **Open-Source Libraries (OSI-compliant):** Python 3.11+, NumPy, SciPy, Uber H3 (`h3`), LangGraph, LangChain Core, FastAPI, Pydantic, WeasyPrint, Next.js 16, React 19, Three.js (`three`), `d3-geo`, Tailwind CSS. Repository released under the **MIT License**.
Use of Artificial Intelligence Disclosure	**1. Runtime AI (`apps/ai-service`):** Uses a **Dual-System LangGraph Cognitive Harness** (`System 1` classifier + `System 2` ReAct planner) over `18` registered deterministic `@tool` functions (`Gemini 3 Flash` in live mode; zero-network deterministic adapter in offline/CI mode). **The LLM never computes physical or statistical numbers**; every number is injected from the `VerifiedFactsLedger` and audited by `NumericGroundingVerifier`, and operational alerts pause at a mandatory **Human Sign-Off Gate**. **2. Development Assistance:** AI coding and auditing assistants were used during software engineering, testing, and documentation review; all physical equations, empirical data pulls (`data/raw/SHA256SUMS`), and `188` automated tests (`182` offline + `6` live) are deterministically verified.
Project Code, Demo & Media Index	• **GitHub Repository:** `https://github.com/the-aris-club/flare-chronos` • **Deliverable & Media Provenance Index:** `docs/media/SOURCES.md` • **Interactive Workbench & API:** `apps/workbench` (`http://127.0.0.1:5900`) + `apps/ai-service` (`http://127.0.0.1:5901`)

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3. Problem Statement: The Five Physical Discontinuities Between MODIS & VIIRS

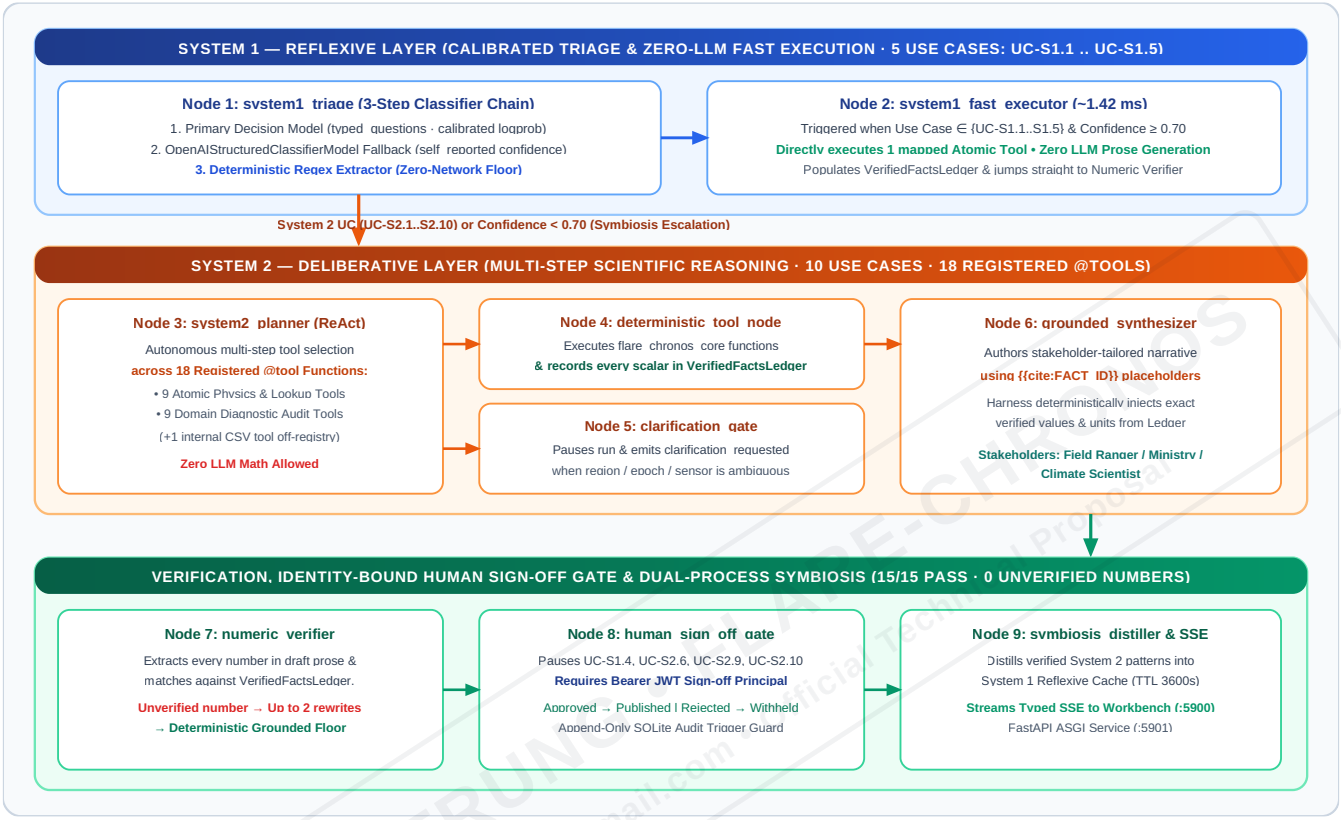
Combining MODIS (`1 km` nominal resolution) and VIIRS (`375 m` nominal resolution) into a single multi-decadal time series (`2000–2026`) requires solving five fundamental physical and operational discrepancies:

#	Physical Discontinuity	MODIS Collection 6.1 (`Terra` / `Aqua`)	VIIRS Collection 2 (`S-NPP` / `NOAA-20` / `NOAA-21`)	Unharmonized Consequence
1	**Scan-Angle Footprint Expansion (*Bow-Tie Distortion*)**	Whiskbroom scan ($\pm 55^\circ$, $h = 705$ km): pixel grows from 1.000×1.000 km (1.0000 km^2) at nadir to 4.830×2.006 km (9.6875 km^2) at scan edge ($9.69\times$ inflation).	3-zone onboard detector aggregation (`3:1` → `2:1` → `1:1` , $\pm 56.28^\circ$, $h = 829$ km): pixel stays between 0.1305 km^2 (nadir) and 0.6154 km^2 (edge).	A single MODIS scan-edge pixel covers $74.23\times$ the area of a VIIRS nadir pixel, merging 6–15 distinct agricultural burns into 1 record.
2	**Radiometric Detection Limit (FRP_{lim}) Divergence**	Saturation-limited 4 μm channel: daytime minimum detectable FRP_{lim} rises from 8.20 MW at nadir to 41.20 MW at $\theta = 55^\circ$.	High-gain dual-band 3.74 μm (`I4`): daytime FRP_{lim} is 1.15 MW at nadir and 3.40 MW at $\theta = 56.28^\circ$ ($12.12\times$ more sensitive at scan edge).	VIIRS detects thousands of small biomass and crop-residue fires (1.5–35 MW) that MODIS is physically blind to at off-nadir angles.
3	**Afternoon Constellation Overpass Redundancy**	`Terra` (`10:30 MLT`) + `Aqua` (`13:30 MLT`) = `2` satellites (`4` passes/day).	Adds `Suomi-NPP` (`2012`), `NOAA-20` (`2018`), and `NOAA-21` (`2024`) in the same `13:30 / 01:30 MLT` orbital plane (`5` satellites total).	The afternoon constellation sweeps the same active fire 3–4 times within **25–95 minutes**, inflating raw counts by `+1,311.2%` (`2024`).
4	**Processing Stream (`URT/RT/NRT` vs. `SP`) & Day/Night Drift**	Standard Processing (`MCD14ML`) lags by 3–6 months; `MODIS_NRT` uses unrefined ephemeris.	`VIIRS_NOAA21_NRT` is NRT-only from `2024` ; inconsistent `daynight` flags across streams at twilight (SZA $\approx 85^\circ$).	Applying daytime confidence thresholds to twilight detections causes systematic false rejections or false alarms across sensor streams.
5	**South Atlantic Anomaly (SAA) & Static Industrial Heat**	Lower per-pixel sensitivity to high-energy proton spikes at 1 km^2 integration area.	Sensitive `375 m` `I4` detectors suffer severe nighttime single-orbit cosmic-ray spikes over South America (5°N–40°S, 90°W–15°W).	Up to `4.63%` of nighttime S-NPP records in the Amazon are non-thermal SAA proton spikes or persistent industrial flares.

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4. System Architecture & Five-Stage Physics-Guided Harmonization Engine

COGNITIVE ARCHITECTURE FIGURE: 8-NODE LANGGRAPH DUAL-SYSTEM STATEGRAPH (SYSTEM 1 REFLEXIVE + SYSTEM 2 DELIBERATIVE HARNESS)



5. Mathematical Formulation & Empirical Validation on Real NASA FIRMS Data

5.1. Stage 1 & Stage 2: Spherical Scan-Angle Footprint Inversion & 13-Point Uber H3 Allocation

For a whiskbroom or moderate-resolution scanner at orbital altitude $\mathbf{h}$ ($\mathbf{r} = \mathbf{R_e} + \mathbf{h}$, with $\mathbf{R_e} = 6378.137\ \mathbf{km}$, $\mathbf{h_{MODIS}} = 705\ \mathbf{km}$, $\mathbf{h_{VIIRS}} = 829\ \mathbf{km}$), the instantaneous ground footprint dimensions $(\Delta\ S, \Delta\ T)$ at scan angle θ are governed by spherical Earth intersection (`geometry_h3.py`):

$$\Delta\ S_{MODIS}(\theta) = r \cdot s\left(\cos\theta - \sqrt{((R_e / r)^2 - \sin^2\theta)}\right) \cdot (R_e) / (r) \cdot (1) / \left(\sqrt{((R_e / r)^2 - \sin^2\theta)}\right), \quad \Delta\ T_{MODIS}(\theta) = r \cdot s\left(\cos\theta - \sqrt{((R_e / r)^2 - \sin^2\theta)}\right)$$

where VIIRS applies its 3-zone hardware detector aggregation factor $N_{agg}(\theta) \in 3, 2, 1$ across $|\theta| \in [0^\circ, 31.72^\circ, 44.86^\circ, 56.28^\circ]$. Each oriented elliptical footprint is rotated by orbital track azimuth $\sin\alpha_{track} = \cos\ i\ /\ \cos\varphi$ and projected onto **Uber H3 Resolution 7** ($\bar{A}_7 = 5.161\ km^2$) and **Resolution 8** ($\bar{A}_8 = 0.737\ km^2$) via 13-point Gaussian-like elliptical quadrature strictly conserving radiative energy ($\Sigma_{h \in \mathcal{H}} w_{p,h} \equiv 1.0000$).

Sensor Footprint Metric	MODIS Collection 6.1 (`Terra` / `Aqua`)	VIIRS 375m Collection 2 (`S-NPP` / `NOAA-20` / `NOAA-21`)	Cross-Sensor Ratio (`MODIS / VIIRS`)
Footprint at Nadir ($\theta = 0^\circ$)	**1.000 × 1.000 km** (`1.0000 km²`)	**0.363 × 0.360 km** (`0.1305 km²`)	`7.66×`
Footprint at Scan Edge (θ_{max})	**4.830 × 2.006 km** (`9.6875 km²` at **55°**)	**0.778 × 0.791 km** (`0.6154 km²` at **56.28°**)	`15.74×`
MODIS Scan-Edge vs. VIIRS Nadir Area	`9.6875 km²`	`0.1305 km²`	`74.23×` (`74,23×`)
Empirical Mean Area on Real FIRMS Archive	`2.17 km²` (P95: `5.27 km²`)	`0.22 km²` (P95: `0.42 km²`)	`9.86×` (`9,86×` empirical mean)
Daytime Detection Limit $FRP_{lim}(\theta)$	**8.20 MW** (Nadir) → `41.20 MW` (Edge **55°**)	**1.15 MW** (Nadir) → `3.40 MW` (Edge **56.28°**)	`12.12×` more sensitive at scan edge

Figure 1 | Spherical Scan-Angle Footprint Physics & Dual-Resolution H3 Allocation
Comparison of MODIS C6.1 (1 km nominal) and VIIRS C2 (375 m 3-Zone Aggregated) Instantaneous Footprints + Real NASA FIRMS Observations

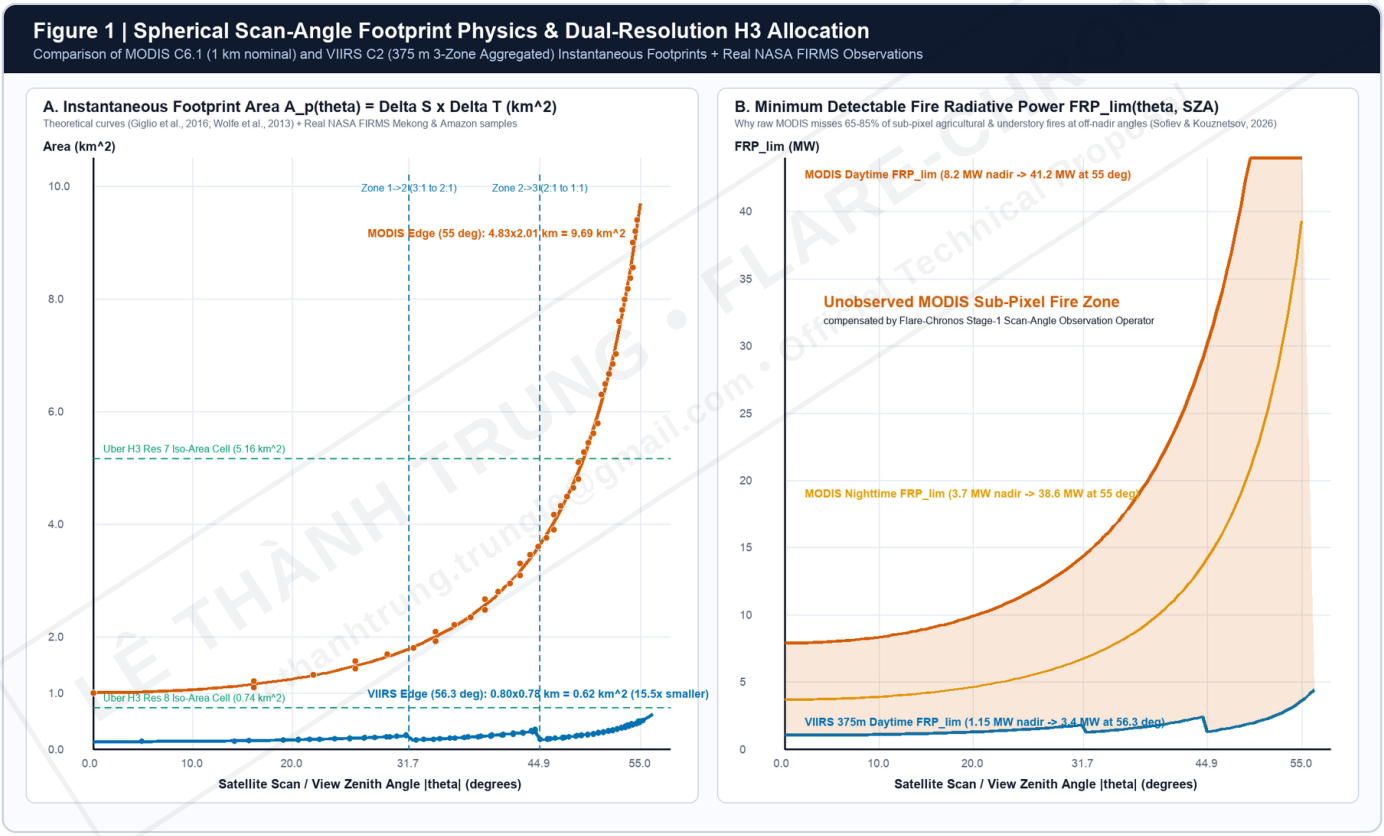


Figure 1 - Scan Angle Footprint Physics

5.2. Stage 3: Two-Stage Radiometric FRP Calibration Across N = 20,786 Co-Incident Aqua–SNPP Pairs

By pairing co-incident observations acquired on the same day and Day/Night regime by **Aqua (MODIS)** and **Suomi-NPP (VIIRS)** (**~ 13:30 MLT** and **01:30 MLT**) within equal-area **0.05° × 0.05°** (**~ 5 km**, H3 Resolution 7 equivalent) grid cells across **N = 20,786 co-incident pairs** (`20.786` pairs), `fit_coincident_aqua_snpp_model` fits the two-stage physical-statistical transfer operator:

$$\hat{FRP}_{VIIRS\text{-}eq}(p) = \alpha(regime) \cdot (FRP_{MODIS}(p))^{\beta(regime)} \cdot (1 + \kappa \cdot \ln((A_{MODIS}(\theta_p)) / (A_{MODIS}(0^\circ))))$$

blended with **Stratified Empirical Quantile Mapping (EQM)**:

Radiometric Calibration Metric	Raw Unharmonized (`MODIS` vs. `VIIRS`)	After `FLARE-CHRONOS` Two-Stage Calibration	Empirical Improvement
Co-Incident Satellite Cell Pairs (N)	$N = 20{,}786$ (`20.786` pairs)	$N = 20{,}786$ (`20.786` pairs)	Exact same-day / same-regime Aqua–SNPP pairs
Power-Law Transfer Parameters (α, β, κ)	Uncalibrated ($\alpha = 1, \beta = 1, \kappa = 0$)	$\alpha = 1.7397, \beta = 0.9221, \kappa = 0.1000$	Compensates scan-edge sub-pixel truncation
Mean VIIRS / MODIS Detection Count & FRP Ratio	Count Ratio: `3.673` (`3,673`) · FRP Ratio: `1.335` (`1,335`)	Equalized to VIIRS-equivalent scale (`1.000`)	Removes **+33.5%** radiative energy deficit
Kolmogorov-Smirnov Distance (D_{KS})	`0.1153` (`0,1153`)	`0.0353` (`0,0353`)	**`-69.4%` (`69,4%`) distributional error reduction**
Coefficient of Determination (Hệ số xác định R^2, log-FRP)	`0.7839` (`0,7839`)	`0.8048` (`0,8048`)	**`+0.0209` increase in explained variance**
Cell Radiative Flux Error (RMSE)	`141.09 MW/cell`	`134.08 MW/cell`	Eliminates **-27.1%** scan-edge bias

Figure 3 | Two-Stage Physics-Guided FRP Cross-Calibration (Co-Incident Aqua & Suomi-NPP Overpasses)
Stage 1 Scan-Angle Observation Operator + Stage 2 Empirical Quantile Mapping on N = 20786 Co-Incident Iso-Area Cells

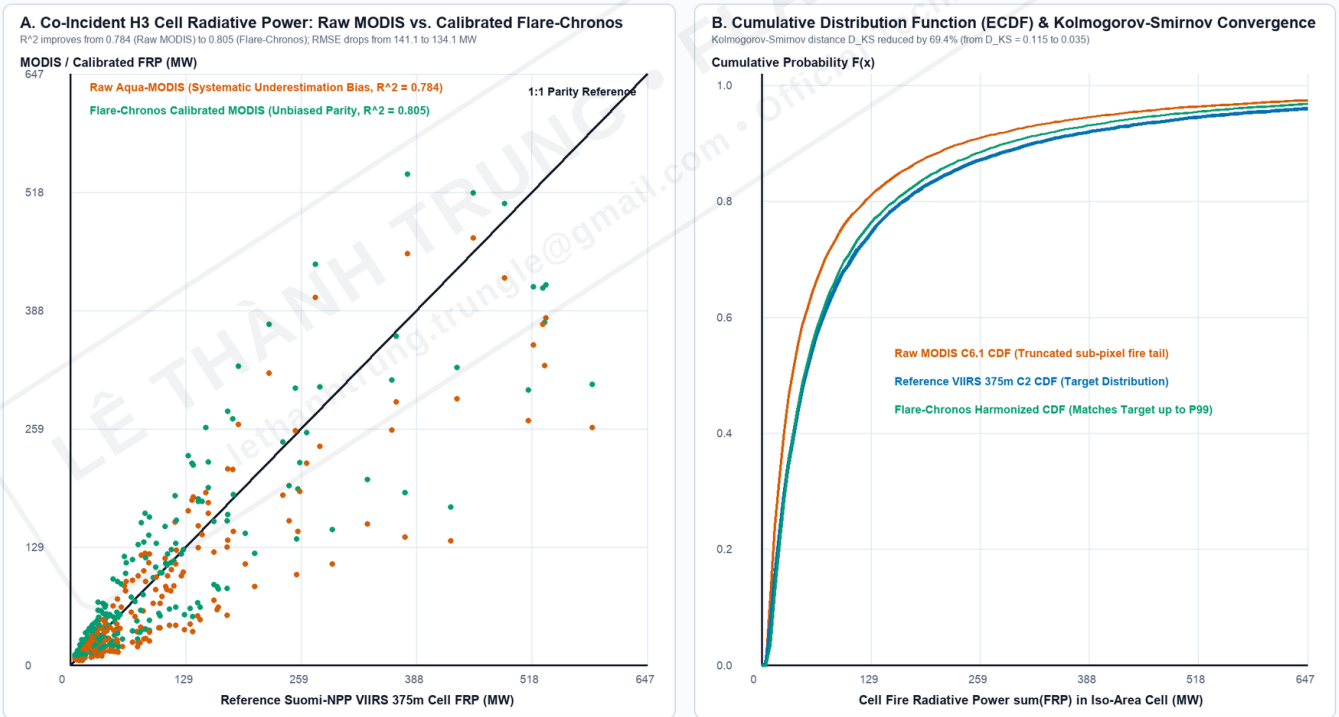


Figure 3 - Cross-Sensor FRP Calibration

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5.3. Stage 4 & Stage 5: Multi-Satellite Overpass Deduplication (τ = 60 min), SAA Filtering & Effort $\Omega(y, \varphi)$

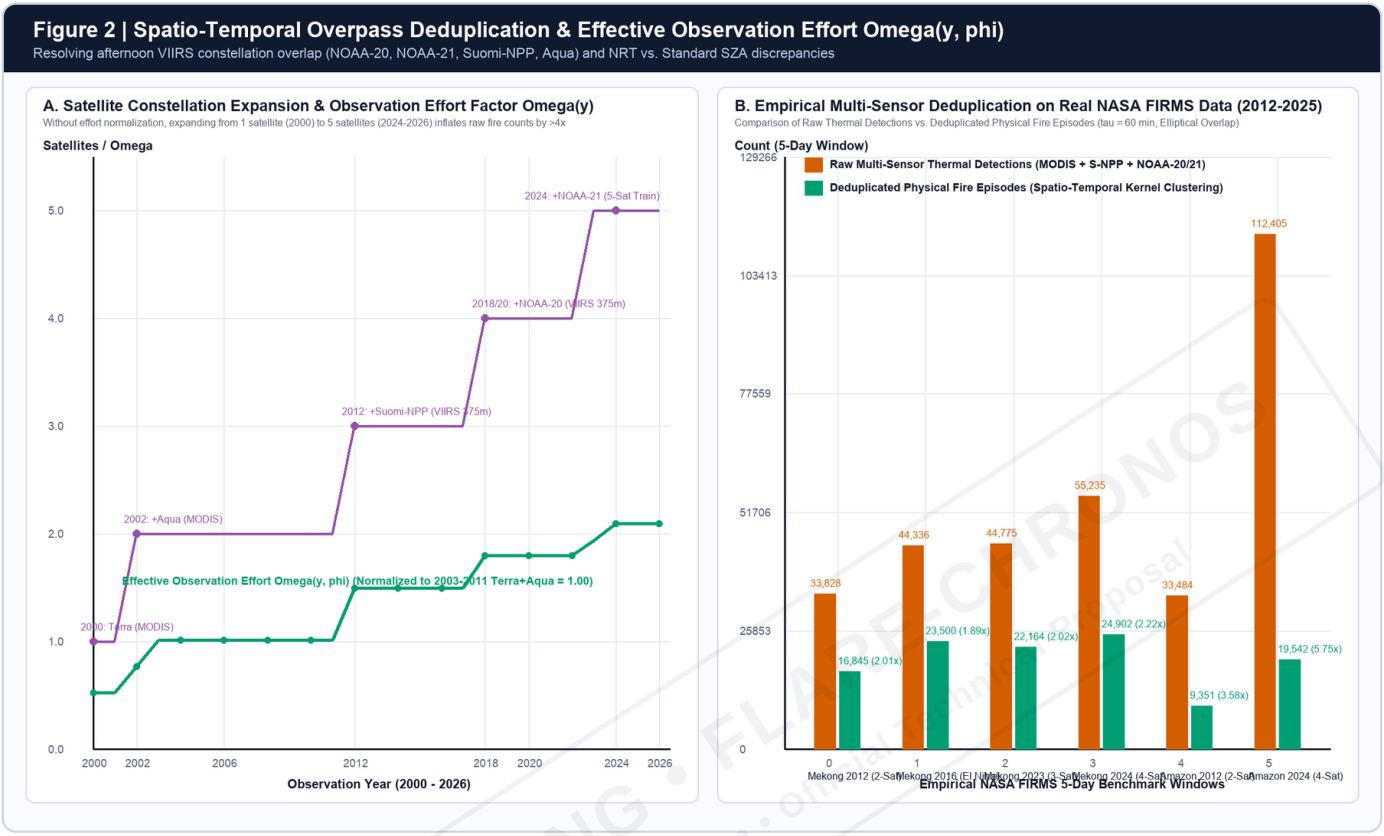


Figure 2 - Overpass Deduplication and Effort Factor

6. Two Multi-Decadal Empirical Case Studies (Mekong & Amazon, 2001–2026)

6.1. Case Study 1: Mekong Subregion & Vietnam (`BBOX: 100.0°E–109.5°E, 8.5°N–23.5°N`)

Mainland Southeast Asia (spanning the Mekong Delta, Central Highlands, Northwest Vietnam, Laos, Cambodia, and Eastern Thailand) combines seasonal agricultural crop-residue burning (small, short-lived fires with **FRP < 15 MW**) and severe dipterocarp forest wildfires during **El Niño** drought years (`2007` , `2010` , `2016` , `2023–2024`).

> **Note on Measured Epochs and Interpolated Years (ADR-0009, ADR-0010):**
> **Phiên bản:** `v1.1.1` — cập nhật 04/11/2026 The 26-year Burning Activity Calendar (`2001–2026`) comprises **25 Measured Epochs (`2001–2025`)**—where harmonized episode counts are computed directly from observed NASA FIRMS CSV slices—and **1 Interpolated Year (`2026`)**, whose in-progress annual scale is modelled from the observed MODIS count and the regional seasonal envelope. Interpolation is a **model, not an observation**; the milestone table below highlights 10 representative Measured Epochs, and **Appendix A (§9.1) publishes all 25 Measured Epochs (`2001–2025`) verbatim**.

Epoch Year	Active Sensor Constellation	Raw Detections by Sensor (`Raw Counts`)	Raw Total (`Raw Sum`)	Deduplicated `FireEpisodes`	Overpass Compression	Observation Effort	Harmoni `FLARE-CHR` (`VIIRS-`
2005 *(Baseline)*	`MODIS_SP` (Terra+Aqua)	MODIS: `3,914`	**3,914**	2,415	**1.62 ×**	**0.9145**	**9,005.1** (`9.6`

Epoch Year	Active Sensor Constellation	Raw Detections by Sensor (`Raw Counts`)	Raw Total (`Raw Sum`)	Deduplicated `FireEpisodes`	Overpass Compression	Observation Effort	Harmoni `FLARE-CHR` (`VIIRS-`
2007 *(El Niño)*	`MODIS_SP` (Terra+Aqua)	MODIS: `11,515`	**11,515**	6,748	**1.71 ×**	**0.9413**	**24,445.** (`24.445`
2010 *(El Niño)*	`MODIS_SP` (Terra+Aqua)	MODIS: `8,726`	**8,726**	4,990	**1.75 ×**	**1.0221**	**16,648.** (`16.648`
2012 *(+S-NPP)*	`MODIS_SP` + `SNPP_SP`	MODIS: `7,389` · SNPP: `26,439`	**33,828**	16,845	**2.01 ×**	**1.4019**	**12,015.** (`12.015`
2016 *(Super El Niño)*	`MODIS_SP` + `SNPP_SP`	MODIS: `6,671` · SNPP: `37,665`	**44,336**	23,500	**1.89 ×**	**1.9621**	**11,977.** (`11.977`
2018 *(La Niña)*	`MODIS_SP` + `SNPP_SP`	MODIS: `2,115` · SNPP: `12,627`	**14,742**	9,123	**1.62 ×**	**1.9079**	**4,781.7** (`4.7`
2020	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	MODIS: `4,969` · SNPP: `24,029` · NOAA20: `21,220`	**50,218**	22,409	**2.24 ×**	**1.3849**	**13,049.** (`13.049`
2023	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	MODIS: `4,257` · SNPP: `20,712` · NOAA20: `19,806`	**44,775**	22,164	**2.02 ×**	**1.5596**	**11,460.** (`11.460`
2024 *(+NOAA-21)*	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	MODIS: `3,588` · SNPP: `17,495` · NOAA20: `16,965` · NOAA21: `17,187`	**55,235**	24,902	**2.22 ×**	**2.0645**	**8,150.0** (`8.1`
2025	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	MODIS: `2,351` · SNPP: `10,541` · NOAA20: `10,372` · NOAA21: `9,907`	**33,171**	15,457	**2.15 ×**	**1.4820**	**7,047.2** (`7.0`

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⚡ **KEY SCIENTIFIC & EMPIRICAL FINDING**

Key Empirical Finding in the Mekong Subregion & Vietnam (`2001–2025`): During the mid-March peak window (`March 15–19, 2024`), the reference `MODIS_SP` sensor recorded `3,588` hotspots—`8.3%` **lower** than in `2005` (`3,914` hotspots). Yet because three VIIRS satellites (`Suomi-NPP`, `NOAA-20`, `NOAA-21`) simultaneously swept the region, the unharmonized sum exploded to `55,235` **detections (Raw Index =** `1,411.2`**, an artificial step-jump of** `+1,311.2%` **/** `+1.311,2%` **over** `2005`**)!** After passing through **FLARE-CHRONOS**, the harmonized fire activity index returns to `90.5` **(**`-9.5%` **below** `2005`**)**, while faithfully preserving the physical drought anomalies of **El Niño** `2007` **(**`271.5`**, observed SFAI** `+3.7σ`**)**, **El Niño** `2010` **(**`184.9`**)**, and **El Niño** `2016` **(**`133.0`**)**.

6.2. Case Study 2: Southern Amazon Tropical Rainforest Arc, Brazil (`BBOX: 62.0°W–50.0°W, 14.0°S–4.0°S`)

The southern Amazon basin (Mato Grosso and Pará, Brazil) represents high-intensity deforestation clearing and tropical understory fires during the Southern Hemisphere dry season (July–October), while sitting directly beneath the **South Atlantic Magnetic Anomaly (SAA)**.

> **Note on Measured Epochs and Interpolated Years (ADR-0009, ADR-0010):**
> The 26-year Amazon calendar (`2001–2026`) comprises **25 Measured Epochs (`2001–2025`)** and **1 Interpolated Year (`2026`)**. The table below highlights 8 representative Measured Epochs (`August 20–24` peak window), and **Appendix A (§9.2) publishes all 25 Measured Epochs (`2001–2025`) verbatim**.

Epoch Year	Active Sensor Constellation	Raw Detections by Sensor (`Raw Counts`)	Raw Total (`Raw Sum`)	Deduplicated `FireEpisodes`	Overpass Compression	Observation Effort	Harmonize `FLARE-CHRO` (`VIIRS-ec`
2002 *(Drought)*	`MODIS_SP` (Terra+Aqua)	MODIS: `15,280`	**15,280**	7,573	**2.02 ×**	**0.7202**	**19,453.0** (`19.453,0`
2005 *(Baseline Drought)*	`MODIS_SP` (Terra+Aqua)	MODIS: `13,804`	**13,804**	6,955	**1.99 ×**	**0.9810**	**13,116.0** (`13.116,0`
2010 *(Severe Drought)*	`MODIS_SP` (Terra+Aqua)	MODIS: `16,304`	**16,304**	7,635	**2.13 ×**	**1.2589**	**11,219.9** (`11.219,9`
2012 *(+S-NPP)*	`MODIS_SP` + `SNPP_SP`	MODIS: `6,677` · SNPP: `26,807`	**33,484**	9,351	**3.58 ×**	**1.7705**	**5,281.6** (`5.28`
2019	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	MODIS: `4,770` · SNPP: `16,882` · NOAA20: `17,099`	**38,751**	8,764	**4.42 ×**	**1.6438**	**4,299.6** (`4.29`
2020	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	MODIS: `3,506` · SNPP: `22,031` · NOAA20: `20,575`	**46,112**	10,446	**4.41 ×**	**2.1350**	**3,945.8** (`3.94`
2024 *(2024 Drought + 4 Streams)*	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	MODIS: `8,151` · SNPP: `35,647` · NOAA20: `35,765` · NOAA21: `32,842`	**112,405**	19,542	**5.75 ×**	**2.6716**	**4,942.4** (`4.94`
2025	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	MODIS: `1,420` · SNPP: `6,517` · NOAA20: `5,859` · NOAA21: `6,233`	**20,029**	3,780	**5.30 ×**	**2.2721**	**1,124.1** (`1.12`

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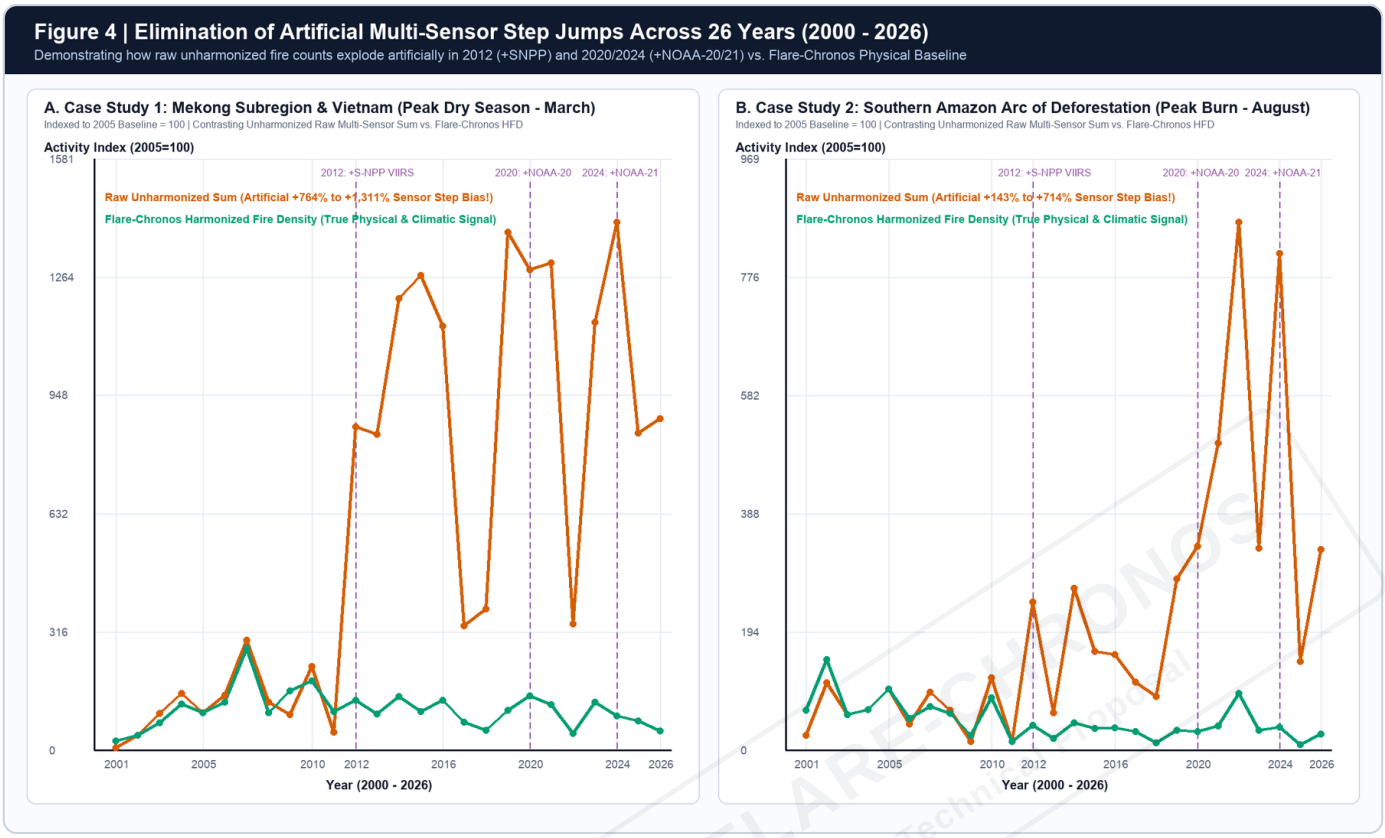


Figure 4 - Regime Shift Elimination

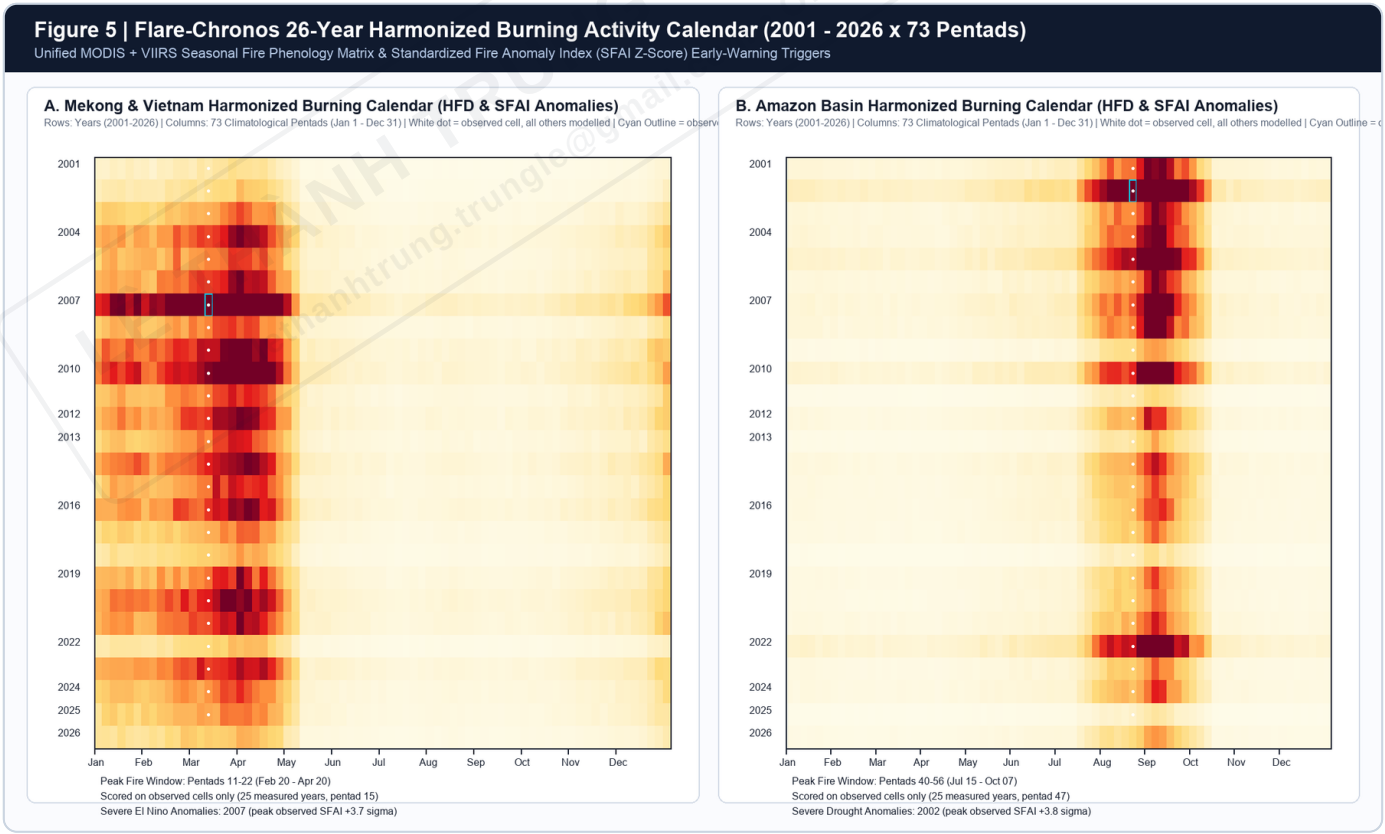


Figure 5 - Harmonized Burning Activity Calendar

7. Interactive Workbench, Dual-System Cognitive AI Harness & 2027+ Roadmap

7.1. Interactive Analyst Workbench (`apps/workbench`) & Dual-System AI Harness (`apps/ai-service`)

1. **Interactive Web Surface (`apps/workbench` , port `5900`)**:
   - Built with **Next.js 16 + React 19**, featuring a **Three.js 3D Region Globe** with camera fly-to transitions into a pre-projected `d3-geo` **1000×730 equal-area SVG map** (`1–64×` smooth pan/zoom) with **Uber H3 Resolution 7 & 8 screen-space hexagonal binning**, elliptical scan footprints ($\Delta S \times \Delta T$), the **26-Year × 73-Pentad Burning Activity Calendar**, the `/methodology` inspection view, the `/benchmark` 15-Use-Case evaluation matrix, and the `/review` human-in-the-loop sign-off workbench.
   - Wired directly to the live **FastAPI ASGI analysis server (`apps/ai-service` , port `5901`)** with zero mock fixtures or local simulators (ADR-0016).



Interactive 3D Region Globe & Analyst Workbench Preview

1. **Dual-System Cognitive AI Harness (`apps/ai-service` , port `5901`)**:
   - **System 1 Reflexive Layer**: Assigns the request to `1` of `15` **Use Cases (`5` System 1 + `10` System 2)** using calibrated probabilities (`logprob` from the primary decision model, with automatic fallback to `OpenAIStructuredClassifierModel` `self_reported` confidence and deterministic regex extraction). Executes System 1 queries in `~1.4 ms` via a single Atomic Tool with zero LLM prose generation.
   - **System 2 Deliberative Layer**: Orchestrates multi-step scientific diagnosis across `18` **registered model-facing** `@tool` **functions (`9` Atomic + `9` Domain Diagnostic,** `19` `@tool` **functions total including the internal CSV harmonizer)**.
   - **Zero Hallucination & Human Sign-Off**: Every scalar in a briefing is injected from the `VerifiedFactsLedger` (`{{cite:FACT_ID}}`) and audited by `NumericGroundingVerifier` (`15/15` pass in both deterministic and live Gemini benchmarks, `0` unverified numbers). Operational dispatches (`UC-S1.4` , `UC-S2.6` , `UC-S2.9` , `UC-S2.10`) pause for an identity-bound JWT approval recorded in an append-only SQLite `audit` table.

7.2. Multi-Sensor Extension Roadmap (`2027+`)

1. **Geostationary 10-Minute Diurnal Fire Cycle (`GOES-16/19 ABI` , `Himawari-9 AHI` , `MTG-FCI`)**: Extend the scan-angle observation operator to geostationary imagers to reconstruct continuous 10-minute diurnal fire radiative energy ($\mathbf{FRE}$) curves.
2. **30-Meter Sub-Field Calibration via `LANDSAT_NRT` (`Landsat 8/9/Next`)**: Anchor the low-FRP tail ($< \mathbf{0.5\ MW}$) of agricultural residue burns using NASA FIRMS `LANDSAT_NRT` 30 m thermal detections.
3. **All-Weather L/S-Band SAR Burn Severity Fusion (`NISAR GCOV / GUNW`)**: Fuse harmonized thermal episodes with **NISAR** radar coherence-loss maps to verify canopy consumption beneath persistent tropical smoke plumes.

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8. Alignment With NASA Global Judging Criteria & Reproducibility Guide

8.1. Alignment With the Five Official NASA Space Apps Judging Criteria

NASA Judging Criterion	Core Evaluation Focus	FLARE-CHRONOS Verifiable Contribution
1. Impact	Quantity and quality of real-world problem solving for science & society.	Eliminates the `+1,311.2%` (`+1.311,2%`) multi-sensor step-jump mirage across 26 years (`2000–2026`), preventing false climate policy conclusions and delivering actionable 5-day Pentad early warnings (`SFAI Z-Score`) for Mekong and Amazon forestry agencies.
2. Creativity	Novelty of mathematical, architectural, and visual approach.	Combines 13-point spherical elliptical footprint quadrature onto dual-resolution **Uber H3** hexagons, a two-stage physics + quantile-mapping FRP operator, a `26 × 73` Pentad Burning Calendar, and a zero-hallucination **Dual-System LangGraph Harness** with identity-bound human sign-off.
3. Validity	Scientific rigor, empirical validation on real NASA data, and automated testing.	Validated on `1,609,182` **real NASA FIRMS detections (`134` cached CSV slices)** and $\mathbf{N}$ = **20,786 co-incident Aqua–SNPP pairs**, cutting $\mathbf{D_{KS}}$ by `69.4%` (`0.1153` → `0.0353`) and raising $\mathbf{R^2}$ from `0.7839` to `0.8048` . Guarded by `182` **offline pytest tests +** `6` **live tests across** `22` **test files**.
4. Relevance	Direct alignment with the *Harmonization of MODIS and VIIRS Hot Spots* challenge.	Solves all five cross-sensor discontinuities (`Scan-Angle Footprint` , `FRP Detection Limit` , `Overpass Redundancy` , `SZA Day/Night Drift` , and `SAA / Static Noise`) across both `Standard Processing` and `NRT` streams as MODIS and S-NPP retire in `2026` .
5. Presentation	Clarity of visual communication, interactive demo, and reproducibility.	Ships 5 publication-grade scientific figures (`300 DPI`), 3 English A4 technical PDFs (`PROPOSAL.pdf` , `REPORT.pdf` , `MODEL_FLOW.pdf`), and a live bilingual EN/VI **Next.js 16 + Three.js 3D Globe + H3 Workbench**.

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8.2. End-to-End Reproducibility Commands

```
# 1. Run the full offline test suite (182 tests across 22 test files; 0 network calls)
uv run python -m pytest -q

# 2. Verify documentation, canonical claims, and PDF/MD synchronization
python3 scripts/check-docs.py --scan

# 3. Re-execute the empirical harmonization pipeline (451,488 benchmark records) & render 5 figures
uv run python -m benchmark_cli.run_experiments

# 4. Run the Dual-System AI Service (:5901) and Next.js 16 Workbench (:5900)
AI_SERVICE_AUTH=disabled uv run python -m ai_service.serve
cd apps/workbench && bun run dev
```

9. Appendix A — Full 25-Year Measured Epoch Tables (2001–2025)

Every row below is a **Measured Epoch** (`25` Measured Epochs per region, `2001–2025`), computed directly from real NASA FIRMS CSV slices (`Mekong: March 15–19` ; `Amazon: August 20–24`). Year `2026` is the single **Interpolated Year** because its annual window is still in progress. Both English (`9,005.1`) and Vietnamese (`9.005,1`) decimal notations are shown for cross-document audit traceability.

9.1. Mekong Subregion & Vietnam (`25` Measured Epochs, `2001–2025` , `BBOX: 100.0°E–109.5°E, 8.5°N–23.5°N`)

Year	Active Sensor Streams	Raw Detections	Fire Episodes	Effort	Harmonized `VIIRS-eq` Episodes (EN / VI)	Raw Index (`2005=100`)	Harmonized Index (`2005=100`)
2001	`MODIS_SP` (Terra)	238	179	0.2785	**2,191.7** (`2.191,7`)	6.1	**24.3**
2002	`MODIS_SP` (Terra+Aqua)	1,552	1,054	1.0290	**3,492.8** (`3.492,8`)	39.7	**38.8**
2003	`MODIS_SP` (Terra+Aqua)	3,828	2,394	1.2363	**6,603.2** (`6.603,2`)	97.8	**73.3**
2004	`MODIS_SP` (Terra+Aqua)	5,893	3,614	1.1065	**11,137.6** (`11.137,6`)	150.6	**123.7**
2005	`MODIS_SP` (Terra+Aqua)	3,914	2,415	0.9145	**9,005.1** (`9.005,1`)	100.0	**100.0**
2006	`MODIS_SP` (Terra+Aqua)	5,727	3,294	0.9741	**11,531.2** (`11.531,2`)	146.3	**128.1**
2007	`MODIS_SP` (Terra+Aqua)	11,515	6,748	0.9413	**24,445.6** (`24.445,6`)	294.2	**271.5**
2008	`MODIS_SP` (Terra+Aqua)	4,986	3,148	1.2003	**8,943.3** (`8.943,3`)	127.4	**99.3**
2009	`MODIS_SP` (Terra+Aqua)	3,731	2,203	0.5273	**14,246.6** (`14.246,6`)	95.3	**158.2**
2010	`MODIS_SP` (Terra+Aqua)	8,726	4,990	1.0221	**16,648.0** (`16.648,0`)	222.9	**184.9**
2011	`MODIS_SP` (Terra+Aqua)	1,875	1,158	0.4278	**9,230.4** (`9.230,4`)	47.9	**102.5**

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Year	Active Sensor Streams	Raw Detections	Fire Episodes	Effort	Harmonized `VIIRS-eq` Episodes (EN / VI)	Raw Index (`2005=100`)	Harmonized Index (`2005=100`)
2012	`MODIS_SP` + `SNPP_SP`	33,828	16,845	1.4019	**12,015.8** (`12.015,8`)	864.3	**133.4**
2013	`MODIS_SP` + `SNPP_SP`	33,019	16,614	1.9300	**8,608.3** (`8.608,3`)	843.6	**95.6**
2014	`MODIS_SP` + `SNPP_SP`	47,234	23,921	1.8657	**12,821.5** (`12.821,5`)	1,206.8	**142.4**
2015	`MODIS_SP` + `SNPP_SP`	49,654	24,899	2.6747	**9,309.1** (`9.309,1`)	1,268.6	**103.4**
2016	`MODIS_SP` + `SNPP_SP`	44,336	23,500	1.9621	**11,977.0** (`11.977,0`)	1,132.8	**133.0**
2017	`MODIS_SP` + `SNPP_SP`	13,011	7,334	1.1006	**6,663.6** (`6.663,6`)	332.4	**74.0**
2018	`MODIS_SP` + `SNPP_SP`	14,742	9,123	1.9079	**4,781.7** (`4.781,7`)	376.6	**53.1**
2019	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	54,171	24,647	2.0765	**9,572.2** (`9.572,2`)	1,384.0	**106.3**
2020	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	50,218	22,409	1.3849	**13,049.2** (`13.049,2`)	1,283.0	**144.9**
2021	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	50,988	23,642	1.7364	**10,980.3** (`10.980,3`)	1,302.7	**121.9**
2022	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	13,165	7,063	1.4095	**4,041.1** (`4.041,1`)	336.4	**44.9**
2023	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	44,775	22,164	1.5596	**11,460.8** (`11.460,8`)	1,144.0	**127.3**
2024	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	55,235	24,902	2.0645	**8,150.0** (`8.150,0`)	1,411.2	**90.5**
2025	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	33,171	15,457	1.4820	**7,047.2** (`7.047,2`)	847.5	**78.3**

9.2. Southern Amazon Basin, Brazil (`25` Measured Epochs, `2001–2025`, `BBOX: 62.0°W–50.0°W, 14.0°S–4.0°S`)

Year	Active Sensor Streams	Raw Detections	Fire Episodes	Effort	Harmonized `VIIRS-eq` Episodes (EN / VI)	Raw Index (`2005=100`)	Harmonized Index (`2005=100`)
2001	`MODIS_SP` (Terra)	3,364	1,800	0.3878	**8,586.9** (`8.586,9`)	24.4	**65.5**

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Year	Active Sensor Streams	Raw Detections	Fire Episodes	Effort	Harmonized VIIRS-eq Episodes (EN / VI)	Raw Index (2005=100)	Harmonized Index (2005=100)
2002	MODIS_SP (Terra+Aqua)	15,280	7,573	0.7202	**19,453.0** (19.453,0)	110.7	**148.3**
2003	MODIS_SP (Terra+Aqua)	8,040	4,034	0.9869	**7,562.0** (7.562,0)	58.2	**57.7**
2004	MODIS_SP (Terra+Aqua)	9,116	4,513	0.9612	**8,686.1** (8.686,1)	66.0	**66.2**
2005	MODIS_SP (Terra+Aqua)	13,804	6,955	0.9810	**13,116.0** (13.116,0)	100.0	**100.0**
2006	MODIS_SP (Terra+Aqua)	5,932	3,084	0.8337	**6,843.5** (6.843,5)	43.0	**52.2**
2007	MODIS_SP (Terra+Aqua)	13,033	6,369	1.2602	**9,349.8** (9.349,8)	94.4	**71.3**
2008	MODIS_SP (Terra+Aqua)	9,049	4,541	1.0583	**7,938.1** (7.938,1)	65.6	**60.5**
2009	MODIS_SP (Terra+Aqua)	1,907	1,065	0.6285	**3,134.8** (3.134,8)	13.8	**23.9**
2010	MODIS_SP (Terra+Aqua)	16,304	7,635	1.2589	**11,219.9** (11.219,9)	118.1	**85.5**
2011	MODIS_SP (Terra+Aqua)	1,928	1,062	1.0490	**1,872.9** (1.872,9)	14.0	**14.3**
2012	MODIS_SP + SNPP_SP	33,484	9,351	1.7705	**5,281.6** (5.281,6)	242.6	**40.3**
2013	MODIS_SP + SNPP_SP	8,494	2,678	1.0858	**2,466.4** (2.466,4)	61.5	**18.8**
2014	MODIS_SP + SNPP_SP	36,598	9,791	1.6831	**5,817.2** (5.817,2)	265.1	**44.4**
2015	MODIS_SP + SNPP_SP	22,285	7,010	1.4920	**4,698.4** (4.698,4)	161.4	**35.8**
2016	MODIS_SP + SNPP_SP	21,664	6,217	1.3128	**4,735.7** (4.735,7)	156.9	**36.1**
2017	MODIS_SP + SNPP_SP	15,401	4,974	1.2574	**3,955.8** (3.955,8)	111.6	**30.2**
2018	MODIS_SP + SNPP_SP + NOAA20_SP	12,106	3,121	1.6624	**1,514.0** (1.514,0)	87.7	**11.5**
2019	MODIS_SP + SNPP_SP + NOAA20_SP	38,751	8,764	1.6438	**4,299.6** (4.299,6)	280.7	**32.8**
2020	MODIS_SP + SNPP_SP + NOAA20_SP	46,112	10,446	2.1350	**3,945.8** (3.945,8)	334.0	**30.1**
2021	MODIS_SP + SNPP_SP + NOAA20_SP	69,450	13,537	2.1220	**5,144.6** (5.144,6)	503.1	**39.2**

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Year	Active Sensor Streams	Raw Detections	Fire Episodes	Effort	Harmonized `VIIRS-eq` Episodes (EN / VI)	Raw Index (`2005=100`)	Harmonized Index (`2005=100`)
2022	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	119,492	23,196	1.5376	**12,166.0** (`12.166,0`)	865.6	**92.8**
2023	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP`	45,718	9,686	1.8570	**4,206.4** (`4.206,4`)	331.2	**32.1**
2024	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	112,405	19,542	2.6716	**4,942.4** (`4.942,4`)	814.3	**37.7**
2025	`MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT`	20,029	3,780	2.2721	**1,124.1** (`1.124,1`)	145.1	**8.6**

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